

HFD 207: TEXTILE AND DESIGN APPLICATION

PRACTICAL HANDOUT: COLOUR THEORY AND COLOUR WHEEL

Practical: Colour Theory and Construction of a Colour Wheel

Medium: Poster colour or watercolour

Paper: A4 cartridge paper

Tools: Pencil, ruler, compass, paintbrushes and mixing palette

1. Practical Objectives

By the end of the practical, the student should be able to:

1. Identify the primary, secondary and tertiary colours.
2. Explain the relationships between colours on the colour wheel.
3. Mix secondary and tertiary colours from primary colours.
4. Demonstrate warm and cool colours.
5. Apply colour harmony principles in textile and surface-design work.

2. Materials and Equipment

Materials/Equipment	Purpose
A4 cartridge paper	Colour-wheel surface
Red, yellow and blue paint	Primary colours
Paintbrushes	Applying colour
Mixing palette/plate	Mixing colours
Water container	Cleaning brushes
Pencil	Construction and labelling
Ruler	Dividing sections
Compass	Drawing the colour wheel
Eraser	Corrections
Black pen/pencil	Labelling

3. Construction of the Colour Wheel

1. Place the A4 paper in **landscape orientation**.
2. Using a compass, draw a large circle approximately **18–20 cm in diameter**.
3. Divide the circle into **12 equal sections**.
4. Mark the positions for the three **primary colours**:
 - Yellow
 - Red
 - Blue
5. Leave one section between each primary colour.
6. Mix the primary colours to produce the **secondary colours**:
 - Yellow + Red = Orange
 - Red + Blue = Violet/Purple
 - Blue + Yellow = Green

7. Place each secondary colour between the two primary colours used to create it.

4. Tertiary Colours

Mix a primary colour with the adjacent secondary colour to produce the six tertiary colours:

- Yellow + Orange = Yellow-Orange
- Red + Orange = Red-Orange
- Red + Violet = Red-Violet
- Blue + Violet = Blue-Violet
- Blue + Green = Blue-Green
- Yellow + Green = Yellow-Green

The completed wheel should therefore contain:

3 Primary + 3 Secondary + 6 Tertiary = 12 Colours

5. Colour Relationships to Identify

Students should identify and label the following:

Complementary colours:

Colours positioned opposite each other on the colour wheel.

Examples:

- Red – Green
- Blue – Orange
- Yellow – Violet

Analogous colours:

Colours positioned next to one another on the colour wheel.

Warm colours:

Generally, colours ranging from yellow through orange to red.

Cool colours:

Generally, colours ranging from green through blue to violet.

6. Practical Exercise

Using the completed colour wheel:

A. Create one complementary colour scheme

Select one pair of complementary colours.

B. Create one analogous colour scheme

Select three adjacent colours.

C. Create one warm colour composition

Use at least three warm colours.

D. Create one cool colour composition

Use at least three cool colours.

Apply each scheme to a small **textile-inspired geometric pattern**.

Expected Outcome

A **12-part colour wheel** showing primary, secondary and tertiary colours, accompanied by correctly identified complementary, analogous, warm and cool colour relationships and simple textile pattern applications.

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Course: HFD 207 – Textile and Design Application